Home Network Design, Configuration and Measurement

Short project report

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Subject	ITFA (course code EF-101)	Instructor	Dr. Fezan Rafique
Submission	Week 13, Monday 9 November 2026	Repository	github.com/syedhussainriaz/home-network-project

1. Objective

To design, configure and investigate a small local-area network so that the physical and logical components that allow devices to communicate are understood. The work has two parts: a simulated network in Cisco Packet Tracer (file *networking_project.pkt*) and measurements on my own real hardware.

2. Network used and topology

Simulated network (Packet Tracer): one router, one switch, one access point, three PCs, one printer, one server and one wireless laptop (7 end devices), all in the network 192.168.1.0/24. **Real hardware:** my own laptop (Realtek RTL8821CE Wi-Fi), my Samsung A17 phone and my home router (192.168.0.1). Real measurements were taken on actual hardware, as required.

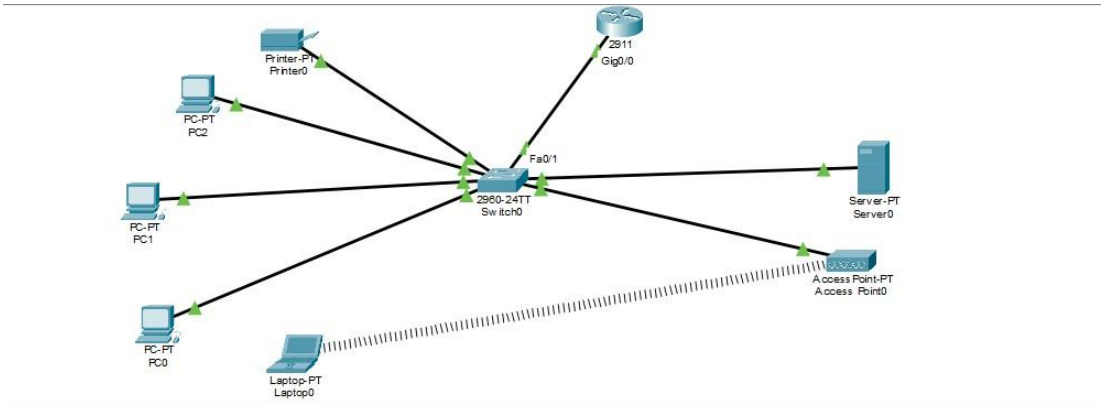


Figure 1: Packet Tracer topology. Solid lines are wired links, the dotted line is the wireless link.

Link	Type	Connected devices
Router0 (Gig0/0) to Switch0 (Fa0/1)	Wired, copper straight-through	Gateway to the LAN
Switch0 to Server0, Printer0, PC0, PC1, PC2, Access Point0	Wired, copper straight-through	Each device has its own port on the switch
Laptop0 to Access Point0	Wireless (Wi-Fi)	Simulated wireless link in Packet Tracer
My laptop and phone to the home router	Wireless (Wi-Fi)	Real network, 2.4 GHz and 5 GHz

3. IP addressing table and device details

Device	IP address	Subnet mask	Default gateway	Assigned by
Router0 (Gig0/0)	192.168.1.1	255.255.255.0	not applicable	Static
Server0 (DNS)	192.168.1.10	255.255.255.0	192.168.1.1	Static
Printer0	192.168.1.11	255.255.255.0	192.168.1.1	Static
PC1	192.168.1.21	255.255.255.0	192.168.1.1	DHCP
PC2	192.168.1.22	255.255.255.0	192.168.1.1	DHCP
PC0	192.168.1.23	255.255.255.0	192.168.1.1	DHCP

Laptop0 (wireless)	192.168.1.24	255.255.255.0	192.168.1.1	DHCP

The router gives DHCP addresses starting at 192.168.1.21 (192.168.1.1 to 192.168.1.20 are excluded for fixed devices) and tells clients to use the DNS server 192.168.1.10. After I reopened the project file, DHCP gave the PCs new addresses in a different order (for example PC0 changed from .21 to .23), so this table uses the addresses seen in the final screenshots.

Device (3 recorded)	MAC address	IP address	Subnet mask	Default gateway
Laptop0 (Packet Tracer, Wireless0)	0001.643B.0C09	192.168.1.24	255.255.255.0	192.168.1.1
Printer0 (Packet Tracer)	0001.C981.741E	192.168.1.11	255.255.255.0	192.168.1.1
My real laptop (Wi-Fi, DHCP)	F8-89-D2-AB-21-B7	192.168.0.105	255.255.255.0	192.168.0.1

The real laptop was given its address by DHCP from the home router (lease about 2 hours).

4. Measurements and results

4.1 Ping and round-trip latency

From	To	Sent / received / lost	Min	Max	Average
PC0 (Packet Tracer)	Router 192.168.1.1	4 / 4 / 0%	0 ms	15 ms	3 ms
PC0 (Packet Tracer)	Server 192.168.1.10	4 / 4 / 0%	0 ms	5 ms	1 ms
PC0 (Packet Tracer)	PC1 192.168.1.21	4 / 4 / 0%	0 ms	0 ms	0 ms
Laptop0 (wireless, Packet Tracer)	Server 192.168.1.10	4 / 4 / 0%	11 ms	45 ms	24 ms
My real laptop	Home router 192.168.0.1	4 / 4 / 0%	1 ms	10 ms	4 ms
My real laptop	google.com	4 / 4 / 0%	17 ms	21 ms	18 ms

The wired PC0 averaged 1 ms to the server, while the wireless Laptop0 averaged 24 ms and varied from 11 to 45 ms, so the wireless link was slower and less steady. Packet Tracer simulates this delay, so those numbers show the idea, not real-world values. The real tests are the real evidence.

4.2 DNS

In Packet Tracer I turned on the DNS service on Server0 and added the record *www.test.com* = 192.168.1.10. From PC0, *nslookup www.test.com* returned 192.168.1.10. On my real laptop, *nslookup google.com* was answered by my home router (192.168.0.1) with the IPv4 address 142.250.186.238 (and an IPv6 address). **Role of DNS:** it works like a phone book, changing a name that people remember into the IP address that computers need.

4.3 Local file transfer and throughput (real hardware)

A 300 MB file (314,572,800 bytes) was sent from my laptop to my Samsung A17 phone using Windows Phone Link. Both devices were on my home Wi-Fi and Bluetooth was not used. My laptop was on the 2.4 GHz band with a strong signal (97% to 100%). The transfer took 398.64 seconds. A first test took about 394 seconds, so the two tests agree.

File size = 314,572,800 bytes x 8 = 2,516.58 megabits. Throughput = 2,516.58 / 398.64 s = **about 6.3 Mbps** (about 0.79 MB/s).

This is far below the Wi-Fi link rate of my laptop (150 Mbps on 2.4 GHz), so the Wi-Fi band was not the limit. The limit was probably the phone or Phone Link, but I did not test the phone separately. A transfer last night of a 295 MB file took 42.63 s (about 58 Mbps), so Phone Link speed changed a lot between sessions.

4.4 Internet speed test and near vs far (real hardware, 2.4 GHz, fast.com)

Condition	Download	Upload	Latency unloaded	Latency loaded
Near the router	12 Mbps	18 Mbps	27 ms	196 ms

Far room (walls between)	9.8 Mbps	17 Mbps	26 ms	473 ms

Both tests were on 2.4 GHz and the Internet provider was CyberInternet (server in Karachi). In the far room download dropped a little and loaded latency rose from 196 ms to 473 ms. I did one test per place, so this is my result in these tests only. My Internet speed also changed from test to test during the two evenings (from about 5 to 16 Mbps), so small differences should not be over-read.

4.5 Default gateway unavailable (Packet Tracer)

Step	Router interface Gig0/0	Test from PC0	Result
1	Up	ping PC1 (192.168.1.21)	Works, 0% loss
2	Shut down	(command shutdown)	Gateway 192.168.1.1 unavailable
3	Down	ping PC1 (192.168.1.21)	Still works, 0% loss
4	Down	ping router (192.168.1.1)	Fails, 100% loss
5	Turned on again (no shutdown)	ping router (192.168.1.1)	Works, 0% loss

When the default gateway is unavailable, devices on the same switch can still talk to each other because the switch does not need the router. But they cannot reach the gateway, so they cannot reach other networks or the Internet. The default gateway is the exit door of the local network.

4.6 Bonus: 2.4 GHz vs 5 GHz Wi-Fi (real hardware)

My router shares one name (HUSSAINIAT) on both bands. I proved the band before each test with *netsh wlan show interfaces*, standing next to the router.

Band	Channel / type	Link rate	Test	Download	Upload	Unloaded	Loaded
5 GHz	153 / 802.11ac	433.3 Mbps	1	13 Mbps	19 Mbps	19 ms	95 ms
5 GHz	153 / 802.11ac	433.3 Mbps	2	16 Mbps	19 Mbps	24 ms	205 ms
2.4 GHz	2 / 802.11n	150 Mbps	1	1.8 Mbps	3.2 Mbps	27 ms	1.6 s
2.4 GHz	2 / 802.11n	150 Mbps	2	1.5 Mbps	9.2 Mbps	29 ms	1.5 s

In this session 5 GHz was clearly better (13 to 16 Mbps against 1.5 to 1.8 Mbps). Last night, in the same place, 2.4 GHz gave 15 Mbps and 5 GHz gave 11 to 12 Mbps, so the 2.4 GHz result changed a lot between the two nights while 5 GHz stayed steadier. A likely reason is interference on the more crowded 2.4 GHz band, but I did not measure interference, so this is only a possible explanation. 5 GHz generally has a shorter range and passes through walls less well (general knowledge, not measured here).

5. Switch, router, access point and network interface card

Device	What it does	In this project
Switch	Connects devices inside one local network and forwards data using MAC addresses.	Switch0 connects the PCs, printer, server, access point and router.
Router	Connects different networks and forwards packets using IP addresses. Often also gives DHCP.	Router0 is the default gateway (192.168.1.1) and DHCP server. My home router connects my devices to the Internet.
Access point	Gives Wi-Fi to wireless devices and joins them to the wired network.	Access Point0 lets Laptop0 join the LAN without a cable.
Network interface card (NIC)	The hardware in a device that connects it to the network. It has a MAC address.	My laptop's Wi-Fi card is a Realtek RTL8821CE with MAC F8-89-D2-AB-21-B7.

6. Required analysis

Latency vs bandwidth. Bandwidth is how much data can be sent per second (my Internet test gave 12 Mbps download). Latency is how long one packet takes to travel (27 ms when the line was quiet). A busy line can keep the same bandwidth but make the delay worse: loaded latency was 196 ms in the same test.

Wi-Fi, distance and interference. Wi-Fi uses radio waves. The signal gets weaker with distance and when it passes through walls, and other Wi-Fi networks on the same channel can interfere. A weaker or noisier signal means lower speed and more delay. In my far-room test loaded latency rose from 196 ms to 473 ms. The 2.4 GHz band is more crowded than 5 GHz, which may explain why it was so slow in my second session.

DHCP vs static addressing. DHCP gives addresses automatically, which is easy and avoids mistakes, but the address can change: after I reopened the file PC0 changed from 192.168.1.21 to 192.168.1.23. Static addresses are typed by hand and stay the same, so I used them for the server (192.168.1.10) and printer (192.168.1.11), which other devices must always find.

Local vs Internet traffic. Local traffic stays inside my network. Ping to my home router averaged 4 ms, while ping to google.com averaged 18 ms because Internet traffic passes through my router, my provider and other networks. Local is not always faster in throughput, though: my local phone transfer (6.3 Mbps) was slower than my Internet download (12 Mbps), because the phone or Phone Link was the limit.

How packets move. To reach a name, the device first asks DNS for the IP address. Data is split into packets. Inside the LAN the switch forwards them using MAC addresses. If the destination is in another network, the device sends them to the default gateway (the router), which forwards them by IP address towards the destination.

7. Safety and ethics

I only configured Packet Tracer and tested my own devices on my own home network. I did not use or disturb university or other people's networks.

8. Conclusion

I built a working network in Packet Tracer with wired and wireless links, static and DHCP addresses and a DNS server, and I tested it with ping, DNS and a gateway failure. On real hardware I measured ping, DNS, a file transfer, Internet speed, near and far Wi-Fi, and 2.4 GHz against 5 GHz. The main lessons were that the gateway is needed only to leave the local network, that Wi-Fi results change with distance and with the band, and that real measurements can vary from test to test, so each result must be reported carefully.

Evidence

All screenshots are in the *screenshots* folder of the repository, the Packet Tracer file is in *packet-tracer*, and the visual schema is shown below and is also in the *schema* folder. Repository: https://github.com/syedhussainriaz/home-network-project

Visual schema of the project

